Pipeline to create analysis ready methylation data from PPMI idat files

Under **../rawdata**, , save the following files into **PPMI/** directory::

- idat files from PPMI repository
- the linkage file **ppmi_140_link_list_20210607.csv** and the covariate file **Participant_Status_16Jul2025.csv** from the PPMI Box folder

The first step is to generate the sample sheets within each of the idat folders. Be sure you have installed pandas in your Python environment. Run the following Python code:

```
import sys
import glob
import pandas as pd
import os



df_link = pd.read_csv('ppmi_140_link_list_20210607.csv')
df_link['PATNO'] = df_link['PATNO'].astype('str')

df_cov = pd.read_csv('Participant_Status_16Jul2025.csv')
df_cov['PATNO'] = df_cov['PATNO'].astype('str')

df_sample_pheno = pd.merge(df_cov,df_link,on='PATNO')[['PATNO','SENTRIXID','POSITION','COHORT_DEFINITIO
phenos = ["Parkinson's Disease","Healthy Control"]
df_sample_pheno['SENTRIXID'] = df_sample_pheno['SENTRIXID'].astype(str)
df_sample_pheno['SAMPLE_GROUP'] = df_sample_pheno['COHORT_DEFINITION']+'_'+df_sample_pheno['EVENT_ID']
print(df_sample_pheno)
idat_files = glob.glob("idats/*")

list_sentrix = []
list_pos = []
for direc in idat_files:
    sentrix = direc.split('/')[-1]
    #print(f'direc {direc} sentrix {sentrix}')
    #sentrix,pos,_ = fname2.split('_')
    #print(f"Filename {sentrix} {pos}")
    df_sample_pheno2 = df_sample_pheno.loc[(df_sample_pheno.SENTRIXID==sentrix) ,:]
    print(f'df_sample_pheno2\n{df_sample_pheno2}')
    if len(df_sample_pheno2) > 0:
        df_sample_pheno2 = df_sample_pheno2.rename(columns={'PATNO':'Sample_Name','SENTRIXID':'Sentrix_
        df_sample_pheno2 = df_sample_pheno2.drop(['COHORT_DEFINITION','EVENT_ID'],axis=1)
        df_sample_pheno2.to_csv(direc+'/sample_sheet.csv',index=False)
```

Next we now will undertake a time-consuming step but memory efficient step of generating various matrices for each set of idat files in each idat folder by running the following R script:

```r
library(ChAMP)

dirs <- list.dirs(path='idats')
for (dir in dirs){
  print(dir)
  csvfile=paste(dir,'sample_sheet.csv',sep='/')
  if (file.exists(csvfile)){
    print(csvfile)
    print('exists')
    myImport<- champ.import(dir,arraytype="EPIC")
    write.table(colnames(myImport$beta),file=paste(dir,'beta_cols.txt',sep='/'),row.names=F,col.names=F
    write.table(rownames(myImport$beta),file=paste(dir,'beta_row.txt',sep='/'),row.names=F,col.names=F,
    write.table(myImport$beta,file=paste(dir,'beta_vals.txt',sep='/'),row.names=F,col.names=F,quote=F,se
    write.table(colnames(myImport$M),file=paste(dir,'M_cols.txt',sep='/'),row.names=F,col.names=F,quote=
    write.table(rownames(myImport$M),file=paste(dir,'M_row.txt',sep='/'),row.names=F,col.names=F,quote=
    write.table(myImport$M,file=paste(dir,'M_vals.txt',sep='/'),row.names=F,col.names=F,quote=F,sep='\t
    write.table(colnames(myImport$pd),file=paste(dir,'pd_cols.txt',sep='/'),row.names=F,col.names=F,quo
    write.table(rownames(myImport$pd),file=paste(dir,'pd_row.txt',sep='/'),row.names=F,col.names=F,quot
    write.table(myImport$pd,file=paste(dir,'pd_vals.txt',sep='/'),row.names=F,col.names=F,quote=F,sep='
    write.table(colnames(myImport$intensity),file=paste(dir,'intensity_cols.txt',sep='/'),row.names=F,c
    write.table(rownames(myImport$intensity),file=paste(dir,'intensity_row.txt',sep='/'),row.names=F,co
    write.table(myImport$intensity,file=paste(dir,'intensity_vals.txt',sep='/'),row.names=F,col.names=F
    write.table(colnames(myImport$detP),file=paste(dir,'detP_cols.txt',sep='/'),row.names=F,col.names=F
    write.table(rownames(myImport$detP),file=paste(dir,'detP_row.txt',sep='/'),row.names=F,col.names=F,
    write.table(myImport$detP,file=paste(dir,'detP_vals.txt',sep='/'),row.names=F,col.names=F,quote=F,se
    write.table(colnames(myImport$beadcount),file=paste(dir,'beadcount_cols.txt',sep='/'),row.names=F,c
    write.table(rownames(myImport$beadcount),file=paste(dir,'beadcount_row.txt',sep='/'),row.names=F,co
    write.table(myImport$beadcount,file=paste(dir,'beadcount_vals.txt',sep='/'),row.names=F,col.names=F
    write.table(colnames(myImport$Meth),file=paste(dir,'Meth_cols.txt',sep='/'),row.names=F,col.names=F
    write.table(rownames(myImport$Meth),file=paste(dir,'Meth_row.txt',sep='/'),row.names=F,col.names=F,
    write.table(myImport$Meth,file=paste(dir,'Meth_vals.txt',sep='/'),row.names=F,col.names=F,quote=F,se
    write.table(colnames(myImport$UnMeth),file=paste(dir,'UnMeth_cols.txt',sep='/'),row.names=F,col.name
    write.table(rownames(myImport$UnMeth),file=paste(dir,'UnMeth_row.txt',sep='/'),row.names=F,col.names
    write.table(myImport$UnMeth,file=paste(dir,'UnMeth_vals.txt',sep='/'),row.names=F,col.names=F,quote=
    #save(myLoad,file=paste(dir,'myLoad.RData',sep='/'))
  }else{
    print('not exists')
  }
  #csv = list.files(path=csvfile)
}
#idatDir='idats/203286230019'
#myLoad <- champ.load(idatDir,arraytype="EPIC")
#save(myLoad,file=paste(idatDir,'myLoad.RData',sep='/'))
```

Next we want to concatenate the results by running this Python script:

```python
import pandas as pd
import glob
import os

matrices=['beta','M','pd','intensity','detP','beadcount','Meth','UnMeth']
samplesheets = ['pd']

def merge_files(pd_file,rows_file,vals_filelist,outfilename):
```

```
    fileout = open(outfilename,'w')
    with open(pd_file,'r') as file:
        file.readline()
        colnames = [line.strip().split('\t')[0] for line in file.readlines()]
        fileout.write('\t'.join(colnames))
    fileout.write('\n')

    with open(rows_file,'r') as file:
        rownames = [line.strip() for line in file.readlines()]
    filehandles = []
    for filename in vals_filelist:
        filehandles.append(open(filename,'r'))
    for index,rowname in enumerate(rownames):
        fileout.write(rowname)
        totalcols=0
        for index2,filehandle in enumerate(filehandles):
            fileout.write('\t')
            linein = filehandle.readline().strip()
            #print(f"index {index2} linein {linein}")
            cols = len(linein.split('\t'))
            totalcols+=cols
            fileout.write(linein)
        fileout.write('\n')
        #print(f"total cols {totalcols}")
        if index % 5000 == 0:
            print(f"Row {index} completed")
    for filehandle in filehandles:
        filehandle.close()
    fileout.close()

def merge_samplesheets(cols_file,vals_filelist,outfilename):
    with open(cols_file,'r') as file:
        colnames = [line.strip() for line in file.readlines()]
    fileout = open(outfilename,'w')
    for index,colname in enumerate(colnames):
        if index>0:
            fileout.write('\t')
        fileout.write(colname)
    fileout.write('\n')
    for filename in vals_filelist:
        with open(filename,'r') as file:
            for line in file:
                linetokens = line.strip().split('\t')
                samplegroupvisit = linetokens[3]
                samplegroup,visit = samplegroupvisit.split('_')
                samplegroup = samplegroup.replace("'","").replace(" ",".")
                fileout.write(linetokens[0]+'_'+visit+'\t'+linetokens[1]+'\t'+linetokens[2]+'\t'+sampleg
                fileout.write('\n')
    fileout.close()



sentrixlist = glob.glob("idats/*")
```

```
def process_datasets(datasets,orientation):
    for matrix in datasets:
        cols_filelist=[]
        vals_filelist=[]
        for index,sentrix in enumerate(sentrixlist):
            if index==0:
                    rows_file=sentrix+'/'+matrix+'_row.txt'
            #print(f"sentrix {index} {sentrix}")
            cols_file=sentrix+'/'+matrix+'_cols.txt'
            cols_filelist.append(cols_file)
            vals_file=sentrix+'/'+matrix+'_vals.txt'
            vals_filelist.append(vals_file)
        outfilename=matrix+'.txt'
        print(f"Output to {outfilename}")
        #print(f"{cols_file} {rows_file} {outfilename} {vals_filelist} ")
        if orientation == 'vertical':
            merge_samplesheets(cols_file,vals_filelist,outfilename)
        elif orientation == 'horizontal':
            merge_files('pd.txt',rows_file,vals_filelist, outfilename)


process_datasets(samplesheets,'vertical')
process_datasets(matrices,'horizontal')
```

Under **PPMI_EPIC/** you should see new files such as **beta.txt** (the Beta values in cpg-major format) and **pd.txt** (A sample sheet that is useful for the ChAMP pipeline). A gzipped copy of **beta.txt** called **ppmi_betas.txt.gz** is located at the PPMI Box folder.